Assignment

1) Hadoop: open source frame work based on java that manages the storage and processing of large amounts of data for application.

Spark: open source distributed processing system used for big data workloads.

- Spark has been found to run 100 times faster in memory & 10 times faster in disk
- Spark uses RAM to cache & process data instead of a file system

2) Limitations of Mango DB: • Joining documents is difficult • Limited data size & Nesting • requires high amount of storage data • Name space is limited to ~123 chars

3) Correlation b/w X & Y

Correlation coefficient: $r = \dfrac{\sum (x-\bar{x})(Y-\bar{Y})}{\sqrt{\sum (x-\bar{x})^2 \sum (Y-\bar{Y})^2}}$

5) Perpendicular: $a \cdot b = 0$

Parallel: $\vec{a}$ should be a scalar multiple of $\vec{b}$ i.e $A = K \cdot B$

6) $A^{-1} = \dfrac{Adj(A)}{|A|}$ Here $A^{-1} = \dfrac{1}{5}\begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ //

4) $A \cdot B = 0$

7) linearly dependent

8) $L(x,y) = x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta$

9) Rank-Nullity Theorem: Let $L: V \to W$ be a linear transformation

Then range of L,

Range (L): It is the image of entire vector space 'V' and finite dimensional

and $\dim(V) = \dim \text{range}(L) + \dim \text{null}(L)$

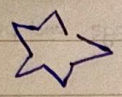
null (L) or Ker (L): set of vectors in 'V' that give zero vector in 'W'

Image (S): Subspace of W.

10) Manhattan norm: $\|x\|_{L_1} = \sum |x_i|$

euclidean norm: $\|x\|_{L_2} = \left(\sum |x_i|^2\right)^{1/2}$

11. Convex function : $f(\lambda x_1 + (1-\lambda) x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$



it is not a convex set

12 ?

13. $|A - \lambda I| = 0$

14. P.S.D $\Leftrightarrow$ $x^T A x \geq 0$

Sym mat is P.D $\Leftrightarrow$ all pivots are +ve.

P.S.D $\Leftrightarrow$ all pivots are non negative

15. $A = PDP^{-1}$ $\quad A^n = (PDP^{-1})^n = \underbrace{PDP^{-1} \cdot P}_{I}\underbrace{DP^{-1} \cdot P}_{I}DP^{-1} \cdots \underbrace{P}_{I}DP^{-1} = PD^nP^{-1}$

16. $A = QDQ^T$

$A = \lambda_1 v_1 v_1^T + \lambda_2 v_2 v_2^T$

17. $A = U \Sigma V^T$

18. $A^+ = (U \Sigma V^T)^+ = U \Sigma^+ V^T$

19. $\|A\|_1 = \max \sum_{i=1} |a_{ij}|$ $\quad \|A\|_\infty = \max \sum_{j=1} |a_{ij}|$

20. Cov mat $\rightarrow$ eigen value $\rightarrow$ normalized eigen vector $\rightarrow$ principal component $\rightarrow$ red^n

21 within class scatter matrix: ~~to~~ S_w

$S_w = S_1 + S_2$

$S_1 = \sum_{x_i \in C_1} (x_i - \mu_1)(x_i - \mu_1)^T$ $\quad S_2 = \sum_{x_i \in C_2} (x_i - \mu_2)(x_i - \mu_2)^T$

22 : right pseudo inverse : $(A^TA)^{-1}A^T$: overdetermined system

23 : MNS : $m < n$: ∞ sol^n : left pseudo inverse : under determined sys

24 : $R^2 = 1 - RV/TV$ $\quad RV : \sum_i |y_i - f(x_i)|^2$

$TV : \sum_i |y_i - \bar{y}|^2$

25. linear rel^n are not sufficient to capture the true pattern.

26. Confusion mat :-

		(Predicted class) Positive	Negative
[Actual class]	+ve	TP	FN
	-ve	FP	TN

27. $\sigma(x) = 1/(1+e^{-x})$

28. logit of $P(x) = \log\left(\frac{P(x)}{1-P(x)}\right)$

29. linear func^n & Sigmoid func^n

30. $F_1 = \frac{2 \times \text{recall} \times \text{Precision}}{\text{Precision} + \text{Recall}}$.